HAO YU

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EDUCATION

Boston University *Sep 2019 - Present*
PhD Student in Computer Science
Advisor: Prof. Margrit Betke

Zhejiang University *Sep 2015 - Jun 2019*
B.E. in Computer Science and Technology
B.A. in English Language and Literature

Chu Kochen Honors College
Overall GPA: 3.95 / 4.0
Advisor: Prof. Jian Wu

PUBLICATIONS

[1] Krishna Kumar Singh, **Hao Yu**, Aron Sarmasi, Gautam Pradeep, Yong Jae Lee, Hide-and-Seek: A Data Augmentation Technique for Weakly-Supervised Localization and Beyond, arXiv preprint arXiv:1811.02545, 2018.

RESEARCH PROJECTS

Measuring and Integrating Facial Expressions and Gaze as Indicators of Engagement and Affect in Tutoring Systems September 2020 - Present
Under the supervision of Prof. Margrit Betke *Boston University*

- Conducted two studies using computer vision techniques to measure students engagement and affective states from their head pose and facial expressions, as they use an online tutoring system, MathSpring.org.
- Present a Head Pose Tutor, which estimates the real-time head direction of students and responds to potential disengagement, and a Facial Expression-Augmented Teacher Dashboard, that identifies students' affective states and provides enhanced information to teachers.
- Preliminary results on our collected video data were encouraging indicating accuracy in detecting head orientation. A usability study was conducted to further evaluate the proposed computer vision software.

Facial Emotion Analysis for Videos of Presidential Candidates September 2019 - Present
Under the supervision of Prof. Margrit Betke *Boston University*

- Collected a database of YouTube videos of candidates for 2020 United States presidential election.
- Annotated the videos with 3 sentiment labels and 7 emotion labels using Crowd sourcing on Amazon Mechanical Turk.
- Created baseline method on the dataset, which uses CNN to extract image features and frame attention technique for video-based facial expression recognition.

Web-based Camera Mouse with Virtual Keyboard September 2019 - May 2020
Under the supervision of Prof. Margrit Betke *Boston University*

- Developed a web-based camera mouse, which is an online assistive service that allows people to control the mouse pointer and type words on a computer with their head or eye movements.
- Used Posenet for online face detection and tracking with normal webcam in the browser.

- Implemented three input options: Dwell time, Reverse Crossing, and Swipe-and-Switch for efficient typing.

Fake Face Video Detection Using Talking Profile January 2019 - April 2019
Under the supervision of Prof. Terence Sim *National University of Singapore*

- Proposed a detection algorithm for recognizing fake face videos generated by face swapping using autoencoders and generative adversarial networks (GANs).
- Used talking profile which consists of multiple types of facial and body motions to identify deepfakes.
- Trained a SVM classifier of talking profiles for each subject using real videos and test using fake videos.

Hide-and-Seek: A Data Augmentation Technique Jul 2018 - Sep 2018
Under the supervision of Prof. Yong Jae Lee *University of California, Davis*

- Applied Hide-and-Seek on improving performance of CNN models of Emotion Recognition, Age Estimation, Gender Estimation and Person Re-identification. The key idea of Hide-and-Seek is to randomly hide patches in a training image, which increases the variety of dataset while preserves spatial alignment.
- Achieved superior performance compared to previous methods for all tasks on the corresponding benchmarks.

Recognition of Invasive Cervical Cancer Based on CNN May 2017 - May 2018
Under the supervision of Prof. Jian Wu *Zhejiang University*

- Developed a visual recognition system of Colposcopy images for auxiliary diagnosis of invasive cervical cancer based on SE-ResNet.
- Trained the network on our dataset of 8556 images and achieved promising performance.

PROFESSIONAL ACTIVITIES

Reviewer for IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI) *2019*

TEACHING

Introduction to Computer Science II (CS 112), Teaching Fellow, Boston University
Instructors: Christine Papadakis-Kanaris and David G. Sullivan *2020 Spring*
Image and Video Computing (CS 585), Teaching Fellow, Boston University
Instructors: Prof. Margrit Betke *2021 Spring*

HONORS AND AWARDS

- First-Class Scholarship for Excellence in Research and Innovation at Zhejing University *2017, 2018*
- Second-Class Scholarship for Outstanding Merits at Zhejiang University *2016, 2018*

RELEVANT COURSEWORK

- **Graduate Coursework:** Machine Learning, Big Data Systems for Data Science, Deep Learning.
- **Undergraduate Coursework:** Computer Vision, Artificial Intelligence, Computer Graphics, Information Retrieval.

TECHNICAL STRENGTHS

Computer Languages	Python, C/C++, Java, Matlab, LaTeX
Deep Learning Frameworks	Tensorflow, PyTorch, Caffe, Keras
Toolkits	OpenCV, OpenGL